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Personalized Energy Consumption Prediction of CAEVs With Personal Driving Cycle Selection

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ABSTRACT For connected and autonomous electric vehicles (CAEVs), drivers must choose a power-efficient route because the energy source of a CAEV is a battery. The driver can find the power-efficient route and reduce the risk of the battery running out if the CAEV navigation system can accurately predict the battery's state of charge (SOC) based on the power consumption prediction for the selected route. As the driver's driving pattern and the vehicle's battery capacity are different for individual vehicles, it is important to predict the power consumption and battery SOC for the selected route. In this paper, the system model of CAEV to collect on-board diagnostics (OBD), OBD data cloud (ODC) to manage the OBD data, and the sequence of message exchange between them is proposed. Further, we propose an algorithm that can learn power consumption based on machine learning (ML) in the ODC. It predicts the battery SOC at the end of a trip using the current SOC when the onboard navigation system requests prediction for a specific route of the CAEV. The algorithm's performance has been evaluated by the OBD dataset of personal vehicles collected in Michigan based on three road types: downtown, urban road, and highway. The results confirmed that the proposed MLR+DNN model could predict battery consumption with an average accuracy of 92%. Additionally, an average accuracy of 94% was obtained in selecting the most similar personal driving cycle (PDC) using hierarchical grey relational analysis (GRA), which is about 15% higher than the accuracy obtained by using the dynamic time warping (DTW) method.

INDEX TERMS Machine learning, energy consumption, personalized prediction, connected and autonomous electric vehicle.

I. INTRODUCTION

A long with electric vehicles (EVs), connected vehicles (CVs), and autonomous vehicles (AVs) are expected to play key roles in the future. CVs are vehicles that can be connected to other vehicles or infrastructure systems through a vehicle communication network [1]. The AVs are vehicles that can be manually driven using various navigation applications. The EV, CV, and AV together form a group of vehicles widely referred to as connected autonomous electric vehicles (CAEVs) [2], [3]. CAEVs help reduce environmental pollution and are equipped with numerous useful features [4], [5]. Therefore, it is essential to develop CAEV-based intelligent vehicles to contribute to the development of advanced

systems, such as smart cities and smart grids [6]. However, factors like price, mileage restrictions, access to charging stations, and charging times require attention, making CAEVs an active research field [7].

The driving range is one of the most important factors in a CAEV [8]. It is the amount of distance covered by the EV in a single charge. Accurate predictions of the driving range of an EV can help the drivers plan the route [9]. Several studies focused on predicting the driving range of an EV [10]–[13]. One such study involved the gradient boosting decision trees (GBDT) machine learning model, wherein the battery electric vehicles (BEV) driving range was accurately predicted. The GBDT model has many additional features that make it more efficient than the regression model [14].

Subsequently, several studies analyzed the factors affecting electric power consumption and then predicted the

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mileage [15], [16]. One such study developed a road-type data-based method that predicted the electric power consumption for a given route. It considered various environmental factors, such as actual driving data that was collected during driving, geographic location of the route, and weather data [17].

After driving, the state of charge (SOC) of a battery needs to be checked. Several studies suggest using a framework or system to collect data of a vehicle during driving [13], [18]. One study developed a simulation model based on the dynamics of EVs. It successfully predicted the changes in power consumption and the battery SOC for any given driving cycle [19]–[22]. However, the developed model was far from capturing reality. In addition, the external noise caused by the actual operation was neglected while calculating the power consumption and the battery SOC values. Therefore, studies have been conducted to collect data from each vehicle and predict power consumption. Similarly, Tseng *et al.* reported that predicting the driver's behavior using the collected data is an important factor [23]. Vatanparvar *et al.* also reported that energy consumption is highly dependent on driving habits [24]. For example, an aggressive driving style is likely to consume more energy than a smooth one.

Factors influencing fuel consumption depending on driving behavior or driving patterns include engine speed and gear shifting behavior, vehicle speed and acceleration [25]. While speed and acceleration have a significant impact on energy consumption, there are differences in power consumption factors related to EVs and ICE vehicles. Since EVs use motors without engines, gear shifts do not occur as often as internal combustion engine (ICE) vehicles. And EVs, unlike ICE vehicles, generate regenerative energy when decelerating, which rather charges the battery [26]. Therefore, not only the acceleration period but also the deceleration period should be considered for power consumption of EVs. Cauwer *et al.* reported a data-based method for predicting energy consumption of EVs, predicting constant motion factor (CMF) related to acceleration, aerodynamic factor (AF) related to speed and power consumption using the Linear Regression Model [27].

Fuzzy logic, Bayesian classifiers, and dynamic time warping (DTW) algorithms were used to analyze and classify the existing driving behaviors [28]–[30]. In particular, the DTW has been widely used to analyze and classify the driver's driving patterns [31]–[33]. It is an effective algorithm used for finding similarities in the time-series data. Another method that can be used to compare the driving patterns or cycles is the grey relational analysis (GRA). GRA is used for calculating the grey relational coefficient (GRC) value for one ideal sequence, while several other sequences are used for comparison. It selects the sequence with the largest GRC value to locate a sequence that is closest to the ideal sequence [34], [35].

In this work, we focus on determining the personal driving cycle (PDC) for a vehicle that is similar to the PDC of any given vehicle based on the power consumption. This is conducted using the GRA along with the global factor

of speed and acceleration. The proposed method uses the hierarchical structure of the local factor in the acceleration and deceleration portions. We also propose a CAEV model that manages the driving information of each vehicle using the data collected from the OBD-II logger and stored in the OBD data cloud (ODC). In addition to learning and predicting the electric power consumption, we propose a method for predicting the battery SOC based on electric power and battery consumption information for each vehicle.

To predict the power consumption and the battery SOC for new routes, we used the DTW and the proposed GRA method, which helped in selecting the driving data that are similar to frequently observed driving behaviors. The battery SOC is predicted using the selected driving data. To increase the prediction accuracy of the power consumption and regenerative braking, we used physics-based equations together with the sensor information gathered from the OBD dataset.

The proposed method can also be used to calculate the battery power per segment using a weighted arithmetic mean to average the sampling times of different sensors. This facilitates predicting the power consumption data obtained using the driving information only.

The contribution of this paper is as follows:

Starting with learning the battery data together with the PDC, the developed method learns and stores the power consumption and power regeneration data. Subsequently, it predicts the power consumption using the PDC while searching for a specific route through navigation. It also predicts the SOC using the battery information of the personal vehicle.

In cases where the personal vehicle driving information for the route found using the navigation system is missing, the steps followed are as mentioned hereafter. Initially, the factors of a hierarchical structure are calculated using the average speed and the average acceleration of a given vehicle, which are obtained using the road type and the time of the route. They are then compared with those obtained using the GRA. After selecting the PDC of another vehicle that is approximately equal to the GRC value, the power consumption is predicted using the PDC. This helps attain power consumption characteristics similar to one's driving pattern.

The rest of this paper is organized as follows. First, Section II presents the public dataset used and essential equations and algorithm for the proposed algorithm. The methodology is described in Section III. Section IV describes the energy consumption prediction results. Finally, the conclusions are summarized in Section V.

II. PRELIMINARIES

A. VEHICLE ENERGY DATASET

The Vehicle Energy Dataset (VED) is a huge dataset collected from November 2017 to November 2018 in Ann Arbor, Michigan, USA. It consists of 383 different private vehicles including ICEs, HEVs, PHEVs, and EVs [36].

This public dataset contains time-series data of fuel, energy, speed, and additional power usage. It includes data collected by the OBD loggers from 264 gasoline vehicles,

TABLE 1. OBD data in the VED.

Data Category	Data Name	Unit	Sampling Time
Driving Data	GPS		
	Latitude	Longitude	deg
	Vehicle Speed (Driving cycle)	km/h	1 sec
	Outside Air Temperature	C	60 sec
	Auxially Power (HVAC)		
	Air-con Power	Heater Power	W
Battery Data	Battery SOC	%	60 sec
	Battery Voltage	V	5 sec
	Battery Current	A	1 sec

92 HEVs, and 27 PHEVs/EVs. Driving scenarios include highways, urban routes with heavy traffic, different driving conditions, and different seasons of a year.

The VED has a benefit over the existing datasets in that it includes travel routes of the individuals and the time-series data of energy consumption.

Table 1 shows the names, units, and sampling times of dynamic time-series data collected by an EV in the VED. It can be seen from the table that the sampling times of the data are not uniform. In the OBD dataset, it can be seen that the Timestamp item and all values are recorded in Table 1. However, it can be seen that some values maintain the previous values or change to the updated values due to different sampling times. Therefore, we can assume that all data are synchronized based on timestamp.

The OBD data are classified into two categories: driving data and battery data. The data which are subject to change while driving such as vehicle velocity and Heating, Ventilation, and Air-Conditioning (HVAC) data belong to the driving data, whereas the data related to the battery such as battery SOC, voltage, and current data belong to the battery data.

B. FORMULAS FOR ELECTRIC VEHICLE MODELING

Assuming that the vehicle is driven on the slope shown in Figure 1 below, the traction force of the vehicle in a specific driving cycle can be expressed as follows.

$$F_t = F_{rd} + F_{ad} + F_{la} \quad (1)$$

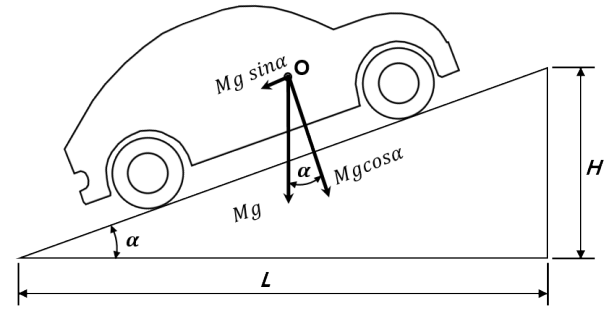
$$F_{rd} = F_f + F_g$$

$$= Mg(f_r \cos(\alpha) + \sin(\alpha)) \quad (2)$$

$$F_{ad} = \frac{1}{2} \rho_a C_D A_f V^2 \quad (3)$$

$$F_{la} = M \delta \frac{dV}{dt} \quad (4)$$

where F_t is traction force of the vehicle, F_{rd} is road resistance, F_f is tire rolling resistance and F_g is grading resistance, F_{ad} is aero dynamic drag, F_{la} is force required to give linear acceleration, V is the speed of the vehicle. M is the total mass of the vehicle, δ is the mass factor that equivalently converts the rotational inertias of rotating components into translational mass.

**FIGURE 1.** Vehicle climbing a grade.

To simplify the calculation, the road angle α is usually replaced by the grade value i . As shown in Figure 1, grade is defined as:

$$i = \frac{H}{L} = \tan(\alpha) \approx \sin(\alpha) \quad (5)$$

When the road angle is small, the road resistance can be simplified as:

$$F_{rd} = Mg(f_r + i) \quad (6)$$

The following are vehicle dynamics formulas (7) and (8) used to model the power consumed or regenerated from the EV battery [26]. These formulas are used in this paper to preprocess data and extract the data features in machine learning modules.

$$P_{batt-out} = \frac{V}{\eta_t \eta_m} F_t + P_{aux} \quad (7)$$

$$P_{batt-in} = \frac{\beta V}{\eta_t \eta_m} F_t$$

$$\text{s.t. } i \text{ or } \frac{dV}{dt} < 0 \quad (8)$$

where P_{aux} is auxiliary power, β is the percentage of the total braking energy that can be regenerated by the electric motor, η_t and η_m are transmission efficiency and motor drive efficiency, respectively.

The equations (9) and (10) are used for calculating battery discharging or charging current based on the power consumed or regenerated from the EV battery, respectively.

Lastly, (11) and (12) represent formulas which are used for predicting a change in SOC based on the battery current.

$$I_{discharge} = \frac{E - \sqrt{E^2 - 4RP_{batt-out}}}{2R} \quad (9)$$

$$I_{charge} = \frac{-E + \sqrt{E^2 + 4RP_{batt-in}}}{2R} \quad (10)$$

$$\int I_{total}(\tau) d\tau = \int (I_{charge}(\tau) + I_{discharge}(\tau)) d\tau \quad (11)$$

$$SOC(t) = SOC(t_0) + \frac{1}{C} \int_{t_0}^t I(\tau) d\tau \times 100 \quad (12)$$

C. DYNAMIC TIME WARPING ALGORITHM

Dynamic time warping (DTW) is an algorithm to find similar patterns in time-series data. The DTW algorithm was originally designed for speech recognition [32].

For example, if there are two signal vectors: $X = \{x_1, x_2, \dots, x_i, \dots, x_m\}$ and $Y = \{y_1, y_2, \dots, y_j, \dots, y_n\}$ on the left and bottom sides of an $m \times n$ grid respectively, each cell in the grid represents the euclidean distance between each point in the signal vector:

$$D(x_i, y_j) = \|x_i - y_j\| \quad (13)$$

Thus, the DTW algorithm can be used to find an optimal alignment of two signal vectors. We used DTW to compare the patterns of the driving cycles of two different personal CAEVs and find a driving cycle which has the most similar pattern.

D. GREY RELATIONAL ANALYSIS

Given n discrete sequences $X_1, X_2, \dots, X_n$, where $X_i = (x_i(1), x_i(2), \dots, x_i(k))$ and $i = 1, 2, \dots, n$, grey relational analysis (GRA) is a useful tool for determining the degree of relationship between these discrete sequences. One of them is designated as the reference sequence (X_0) and represents the ideal situation. It may be used to compute the Grey Relational Coefficient (GRC), which expresses the level of similarity between the reference sequence and the other sequences. The GRC is typically implemented in three steps: data normalization, reference sequence definition and GRC computation and the most preferable sequence is the one with the highest GRC.

The first step, data normalization, is performed according to three possible criteria are “larger-the-better”, “smaller-the-better” and “nominal-the-best”, as shown, respectively, in equations (14)-(16).

$$x_i^*(j) = \frac{x_i(j) - l_j}{u_j - l_j} \quad (14)$$

$$x_i^*(j) = \frac{u_j - x_i(j)}{u_j - l_j} \quad (15)$$

$$x_i^*(j) = 1 - \frac{|x_i(j) - m_j|}{\max\{u_j - m_j, m_j - l_j\}} \quad (16)$$

where $u_j = \max\{x_1(j), x_2(j), \dots, x_n(j)\}$, $l_j = \min\{x_1(j), x_2(j), \dots, x_n(j)\}$, m_j is the target value and $j = 1, 2, \dots, k$.

Next, we define the reference sequence as the one which contains the upper bound, lower bound or moderate bound for the larger-is-better, smaller-is-better or nominal-is-better circumstances, respectively. Then, the deviation sequence Δ_i of the reference sequence X_0 and comparability sequence X_i is computed and, from it, the GRC is computed according to equation (17). The sequence with the largest GRC will be chosen as the one with highest degree of similarity to the reference sequence.

$$GRC_i = \frac{1}{m} \sum_{j=1}^m \frac{\Delta_{\min} + \Delta_{\max}}{\Delta_i + \Delta_{\max}} \quad (17)$$

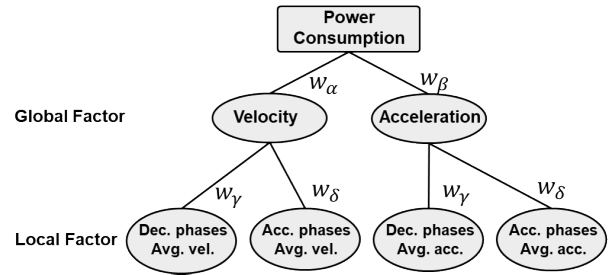


FIGURE 2. Power consumption hierarchy.

where $\Delta_i = |x_0^*(j) - x_i^*(j)|$ is the deviation sequence and $\Delta_{\max} = \max_{(i,j)}(\Delta_i)$, $\Delta_{\min} = \min_{(i,j)}(\Delta_i)$ are the functions to compute the maximum and minimum of the deviation sequence in relation to i and j [34].

In this paper, when there is no personal driving cycle (PDC) generated by the vehicle being driven for the searched route, this paper selects and uses the PDC of another vehicle with the highest degree of similarity in terms of power consumption. The representative PDC, which is the reference for comparison with the PDC of other vehicles, can be selected based on how similar the road types and the times of the day are. The representative PDC can be selected through the DTW-based formula below.

$$\arg \min_i \sum_{j=1}^{\text{all}} D(PDC_i, PDC_j) \quad (18)$$

To compare the method with the GRA technique, weighted factors with a hierarchical structure shown in Figure 2 were used.

In this paper, the reference sequence (X_0) consists of four elements: the two average velocities and two average accelerations of the acceleration and deceleration phases. Then, in the step of data normalization, X_0 becomes the target value (m), so that it can be computed using formula (16).

Also, in Figure 2, the weight of the Global Factor is called global weight (GW), and the weight of the Local Factor is called local weight (LW). Each of them is denoted as follows: $GW = \{w_\alpha, w_\beta\}$, $LW = \{w_\gamma, w_\delta\}$. The final weight for GRA comparison can be obtained by combining the above two weights $W = \{w_1, w_2, w_3, w_4\} = \{w_\alpha w_\gamma, w_\alpha w_\delta, w_\beta w_\gamma, w_\beta w_\delta\}$.

Finally, the weighted GRC can be computed according to equation (19).

$$GRC_i = \frac{1}{\sum_{p=1}^4 w_p |x^*(p) - 1| + 1} \quad (19)$$

III. METHODOLOGY

Figure 3 illustrates the scenario of predicting the vehicle's battery SOC at the end of a trip using an on-board navigation system by the proposed CAEV model and ODC internal algorithms. Upon completion of each trip, the collected OBD data on the CAEV are transmitted to the ODC over a wireless

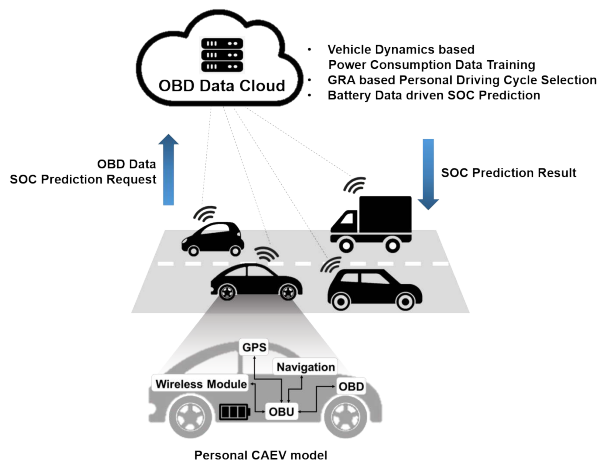


FIGURE 3. The scenario of predicting the EV battery SOC through the CAEV and ODC.

network. The ODC then learns the pattern of power consumption for each vehicle associated with the battery voltage and current of the OBD data over time and saves the result as a model file in the cloud storage.

When the battery SOC prediction is requested via an on-board navigation system of the CAEV, the most similar driving cycle is selected based on the DTW algorithm in the ODC. Then, with a user-specific stored ML model, power consumption and the battery SOC are predicted and delivered back to the CAEV, which shall be displayed on the navigation screen.

Figure 4 shows a detailed data flow diagram between the CAEV and ODC according to three service scenarios.

First, in order for CAEV to obtain SOC prediction results from the ODC, the process of registering the CAEV in the ODC is required. Here, the CAEV sends a registration request message with unique information for each vehicle such as Vehicle Identification Number (VIN). When the ODC receives this message, it creates a database for this vehicle, assigns a unique vehicle ID (Veh ID), and sends a response message to the CAEV. In the generated database, future OBD data, learned power consumption and regenerative ML model of the vehicle are stored.

Next scenario is a process performed every time the CAEV is on a journey. In the CAEV, the end user can be either a human user or a machine user: human if the vehicle is driven manually by a driver, machine if the vehicle drives itself autonomously. When a trip is over, the OBU transmits the collected OBD data information together with the Veh ID previously assigned by the ODC to the DM module of the ODC. The DM module in the ODC preprocesses the received raw OBD data based on the vehicle dynamics formula, and delivers the preprocessed data to the ML module. The ML module learns power consumption or regeneration and stores the results as ML model files.

The last scenario is a case where a route search is requested by the navigation of the CAEV. A human or machine user in the CAEV can ask the navigation for a full route search

to the destination. After the navigation obtains detailed route information, it delivers it to the OBU. Upon receiving this information, the OBU transmits the route data along with the Veh Id in the SOC prediction request message to the DM module of the ODC. The DM module retrieves all the personal driving data with the same GPS locations based on the detailed route data. Among the results, the most similar in terms of power consumption, if not the same, personal driving data will be selected by the DCS module. It adopts the DTW algorithm to calculate the DTW distances between the driving cycles to evaluate the similarity. If the most similar driving cycle is found, the DCS module will forward it to the ML module.

The ML module predicts the power consumption for the entire route from the previously learned model with the driving cycle, current ambient temperature and HVAC power consumption as input data.

The BSP module infers the battery capacity, voltage, and resistance values based on the most recent battery data received from the DM module. With the envisioned total amount of power consumption, it predicts the change in the battery SOC.

This prediction result is transmitted back to the navigation system of the CAEV by the OBU and displayed to the user. If there are is than one candidate route to the destination, the aforementioned procedure is repeated for each route to obtain the prediction results.

Figure 5 shows how the ML model is trained and the SOC is predicted in the ML module and BSP module of the ODC.

Figure 5(a) shows what happens in the ODC when the OBD data collecting and learning process of Figure 4 is performed. When a user transmits OBD data after driving, the DM module of the ODC stores these files in the database. To facilitate the learning process in the ML module, it pre-processes the data split into two modes of the CAEV: the power consumption mode and power regeneration mode. After learning from each data, each of the ML models will be stored in the database again. Figure 5(b) and (c) illustrate the procedures performed by the ODC during the SOC prediction process triggered by the navigation of the CAEV. As shown in Figure 5(b), the DM module searches the existing driving data for the same route as the route searched by the driver using the navigation. Note that the term driving data, as shown in Table 1, refers to the data only directly related to driving among the OBD data, where the battery data are excluded. If one or more driving data are found by the DM module, the most similar data in terms of power consumption to the PDC (i.e., vehicle velocity data over time) are chosen and delivered to the ML module. Based on the trained model as shown in Figure 5(a), the ML module predicts power consumption for the entire route. In the end, the BSP module predicts the battery SOC for the selected route using the battery capacity and initial SOC value of the most recent battery data from the user.

Algorithm 1 is an algorithm showing how the ML module performs preprocessing, classifies power consumption and

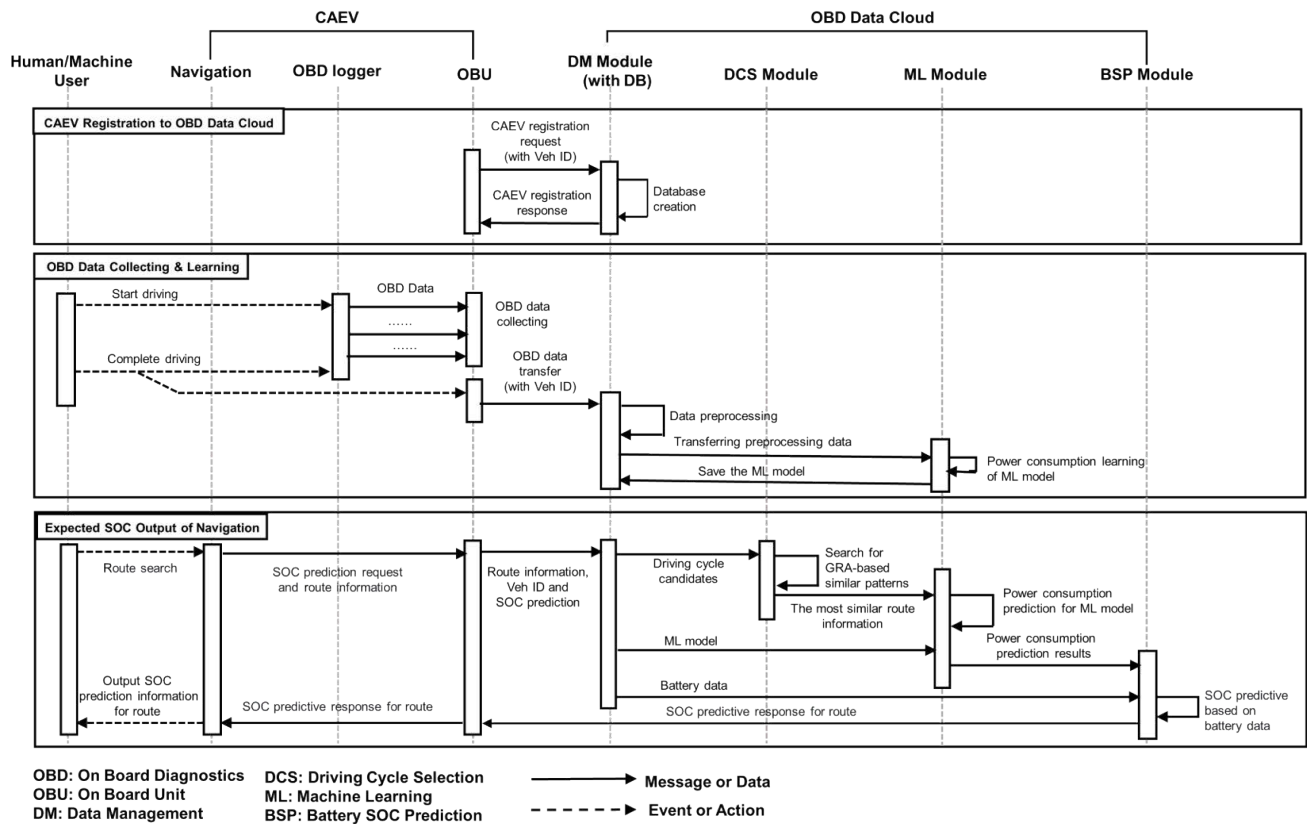


FIGURE 4. Data flow diagram between CAEV and ODC.

regeneration, and learns from the data. In the DM module, the value of $X(i, t)$ is obtained by recalculating the values using the vehicle dynamics (VD), auxially power and environment features as shown in Table 2 with the received Veh ID and OBD data.

Since the EV power consumption can be linearly expressed as the following formula derived from the vehicle dynamics equations (7) and (8), we included $v \times i$, $v \times a$, v^3 as features, besides v and a .

$$P_{EV} = C_1 v i + C_2 v a + C_3 v^3 + P_{aux} \quad (20)$$

Also, as in the previous data driven power consumption prediction study, we included RoadType, TimeOfDay, and Outside Air Temperature as features.

$X(i, t)$ represents $X(i)$ in timestamp t , and $X(i)$ is as follows according to the value of i .

All $X(i)$ values can be taken directly from the raw OBD dataset, except for $X(2)$, $X(7)$ and $X(8)$, which may be computed as follows. First, since i in $X(2)$ is proportional to H by formula (5), it is replaced with the value of H , which was calculated using the difference in height that occurs whenever the car moves. After obtaining the elevation value at each timestamp using the GPS information and the elevation API of Google Maps [37], an approximate value of H could be obtained through the difference from the previous value. $X(7)$ is the Road Type feature, and it was divided into three

TABLE 2. Input variable $X(i)$ of ML module.

$X(i)$	Category	Feature
$X(0)$	Vehicle Dynamics	v
$X(1)$		a
$X(2)$		$v \times i$
$X(3)$		v^3
$X(4)$		$v \times a$
$X(5)$	Auxially Power	Heater Power
$X(6)$		Air-con Power
$X(7)$	Environment	Road Type (Downtown/Urbanroad/Highway)
$X(8)$		Time of Day (0-24)
$X(9)$		Outside Air Temperature (C)

categories, namely, Downtown, Urban Road, and Highway, as in the previous study under the assumption that the type of road could be identified through the current map and GPS location information. Finally, $X(8)$ represents the hour of the day and may assume one of the twenty-four possible values. For example, for 00:20 and 18:30, $X(8)$ will assume the values of 0 and 18, respectively.

$Y(t)$ is the value of $P(t)$, and the value of $P(t)$ is calculated as $V(t) * I(t)$ from the battery data. This way, $X(i, t)$ and $Y(t)$ can be obtained.

However, it is not possible to learn directly from these data. This is because, as shown in Table 1, the sampling

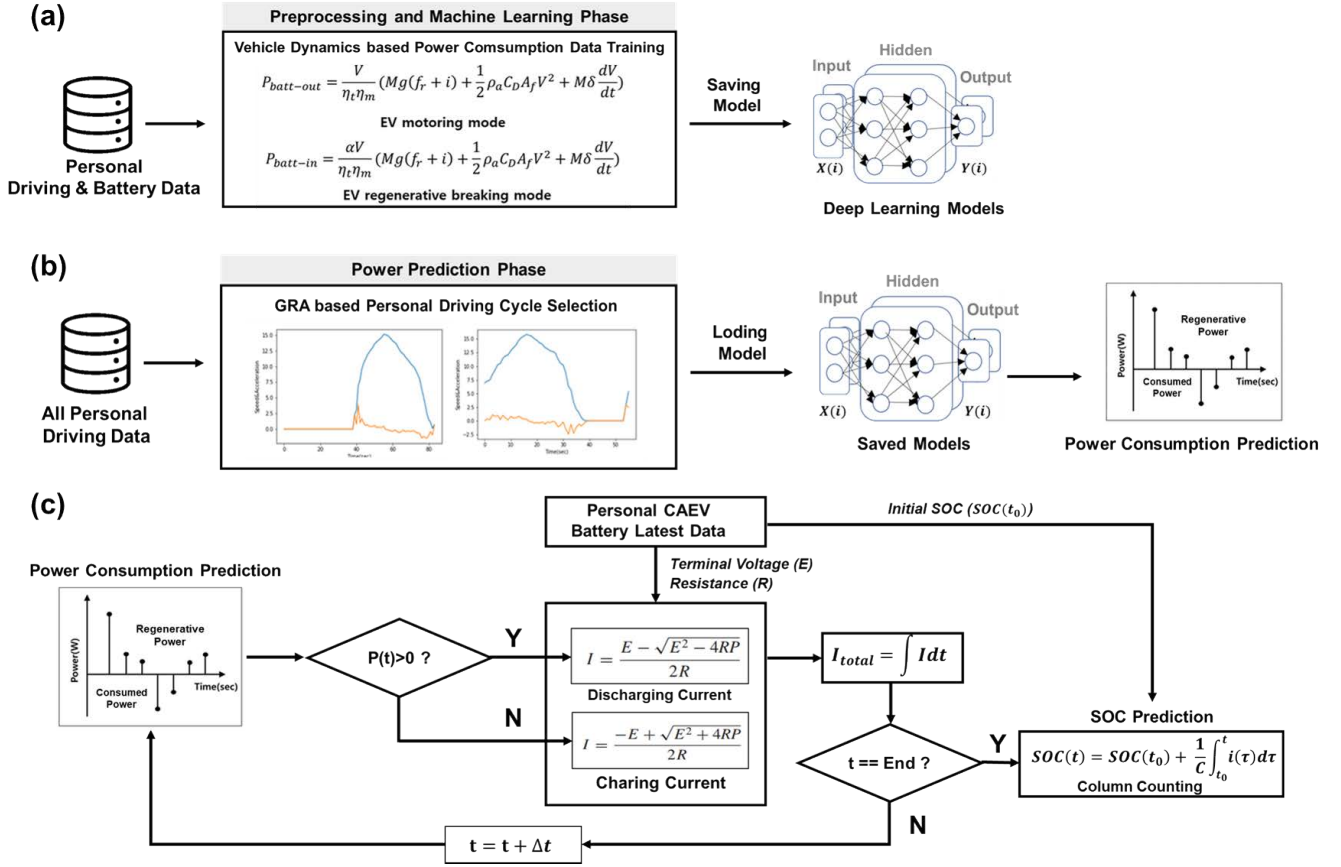


FIGURE 5. Personalized data learning and SOC prediction in ODC.

periods of the OBD data are different. Thus, for data with different timestamp values but with the same $P(t)$, data are reconstructed with an arithmetic weighted mean using $X(i, t)$ data belonging to the interval.

Assuming that T_n represents the n_{th} timestamp, $T_{a,b}$ is defined as follows:

$$T_{a,b} = T_b - T_a \quad (21)$$

where $T_{a,b}$ represents the time between a_{th} timestamp and b_{th} timestamp.

The following equation (22) represents a general weighted arithmetic mean, and (23) represents a weighted arithmetic mean obtained by assigning a weight over time in the j section.

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (22)$$

$$\begin{aligned} X(i, j) &= \frac{\sum_{k=1}^{n-1} T_{k,k+1} X(i, T_k)}{\sum_{k=1}^{n-1} T_{k,k+1}} \\ &= \frac{\sum_{k=1}^{n-1} T_{k,k+1} X(i, T_k)}{\sum_{k=1}^{n-1} T_{1,n}} \end{aligned} \quad (23)$$

In addition, it is divided into data $X_{pc}(i, j)$ for power consumption and data $X_{pr}(i, j)$ for power regeneration according

to the $P(t)$ value as follows:

$$\begin{aligned} X_{pc}(i, j) &= X(i, j) \\ \text{s.t. } P(t) &< 0, t \in j \text{ section} \end{aligned} \quad (24)$$

$$\begin{aligned} X_{pr}(i, j) &= X(i, j) \\ \text{s.t. } P(t) &> 0, t \in j \text{ section} \end{aligned} \quad (25)$$

Next, the $X_{pc}(i, j)$ and $X_{pr}(i, j)$ data are learned by the ML module and stored as $M_{pc}(Veh_id)$ and $M_{pr}(Veh_id)$ according to Veh_id and mode, respectively.

In addition, $X_{pc}(i, j)$ and $X_{pr}(i, j)$ data perform multi-variable linear regression (MLR) based on vehicle dynamics equations as shown in Algorithm 1 (from line 19 to line 27), enabling more accurate prediction by removing data with mean absolute percentage error (MAPE) greater than the predetermined value of α (%). In this paper, the values of N and α are set to 3 and 100 by default, respectively.

Algorithm 2 shows the data generation process for prediction in ODC when there is a request for SOC prediction after driving for a specific route in CAEV navigation. As shown in the formulas (26), (27), data $\hat{X}_{pc}(i, j)$ and $\hat{X}_{pr}(i, j)$ for power consumption and regenerative prediction are generated according to the conditions of acceleration ($a(t)$), respectively.

Algorithm 1 Preprocessing and Training Sequence of the OBD Dataset in the ML Module**Input:** $(X(i, t), Veh_{id})$

```

1: while  $t \neq \text{end}$  do
2:   if  $P(t) < 0$  then
3:     if  $P(t_{curr}) == P(t_{next})$  then
4:        $X_{pc}(i, j_{pc}) \leftarrow \text{Create PC dataset by WAM}$ 
5:        $t \leftarrow \text{next\_timestamp}$ 
6:     else
7:        $j_{pc} \leftarrow j_{pc} + 1$ 
8:     end if
9:   else
10:    if  $P(t_{curr}) == P(t_{next})$  then
11:       $X_{pr}(i, j_{pr}) \leftarrow \text{Create PR dataset by WAM}$ 
12:       $t \leftarrow \text{next\_timestamp}$ 
13:    else
14:       $j_{pr} \leftarrow j_{pr} + 1$ 
15:    end if
16:  end if
17: end while
18: Remove anomalous data
19:  $n \leftarrow 0$ 
20: while  $n < N$  do
21:   Multi-variable linear regression
22:   Evaluate each training data with MAPE
23:   if  $\text{MAPE} > \alpha$  then
24:     Remove data from data set
25:   end if
26:    $n \leftarrow n + 1$ 
27: end while
28: if  $P(t) < 0$  then
29:   Training PC data
30:   Save ML model for PC
31:   return  $M_{pc}(veh\_id)$ 
32: else
33:   Training PR data
34:   Save ML model for PR
35:   return  $M_{pr}(veh\_id)$ 
36: end if
Output:  $M_{pc}$  or  $M_{pr}$ 

```

The generated prediction data is then passed through the process as shown in Figure 5(b), (c) to predict the final battery SOC, and the result is transferred to the user through navigation.

$$\hat{X}_{pc}(i, j) = \hat{X}(i, j) \quad \text{s.t. } a(t) \geq 0, t \in j \text{ section} \quad (26)$$

$$\hat{X}_{pr}(i, j) = \hat{X}(i, j) \quad \text{s.t. } a(t) < 0, t \in j \text{ section} \quad (27)$$

The method described thus far considers a PDC for the specific route searched in the navigation system generated by the vehicle being driven. However, it cannot be applied when such data is not available and, in this case, a new procedure

Algorithm 2 Creating a Data Set for Prediction Based on Driving Route Information on a Specific Route in ODC**Input:** $(RoadType, TimeOfDay)$

```

1: PDC  $\leftarrow (RoadType, TimeOfDay)$ 
2:  $\hat{X} \leftarrow []$ 
3: while  $t \neq \text{end}$  do
4:   if  $a(t_{curr}) \geq 0$  then
5:     if  $a(t_{next}) == 0$  then
6:        $j_{pc} = j_{pc} + 1$ 
7:     else
8:        $\hat{X}_{pc}(i, j_{pc}) \leftarrow \text{Create dataset for}$ 
          PC prediction by WAM
9:        $\hat{X}.\text{append}(\hat{X}_{pc}(i, j_{pc}))$ 
10:    end if
11:   end if
12:   else
13:     if  $a(t_{next}) == 0$  then
14:        $j_{pr} = j_{pr} + 1$ 
15:     else
16:        $\hat{X}_{pr}(i, j_{pr}) \leftarrow \text{Create dataset for}$ 
          PR prediction by WAM
17:        $\hat{X}.\text{append}(\hat{X}_{pr}(i, j_{pr}))$ 
18:     end if
19:   end if
20: end while
21: return  $\hat{X}$ 
Output:  $\hat{X}$ 

```

is proposed. First, by analyzing the inputs RoadType and TimeOfDay, the potentially most similar PDC to the searched route is selected as the representative PDC, which may be computed by equation (18).

After the selection of the representative PDC, the GRA is applied to compare the candidate PDCs of other vehicles with the representative PDC in order to select the most similar one in terms of power consumption. Particularly, for the GRA comparison, the four output values from Algorithm 3 (average speeds and accelerations) and the GRC computed by equation (19) are needed.

Finally, Figure 6 shows how the proposed algorithm predicts when a specific route to a destination is found in the CAEV navigation system, regardless whether or not there is a PDC of the own vehicle in the ODC DB.

IV. PERFORMANCE EVALUATION**A. HYPERPARAMETER SETTING OF DNN MODEL IN ML MODULE**

As shown in Figure 5, our algorithm has two major tasks: predict the power consumption through the ML model and predict the SOC of the battery accordingly.

We first compare the performance of different hyperparameters in the ML module through MAPE metric, as shown in equation (28).

$$\text{MAPE}(\%) = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{x}_i - x_i}{x_i} \right| \times 100 \quad (28)$$

Algorithm 3 Average Velocity and Acceleration Computation for the GRA-Based Method**Input:** PDC

```

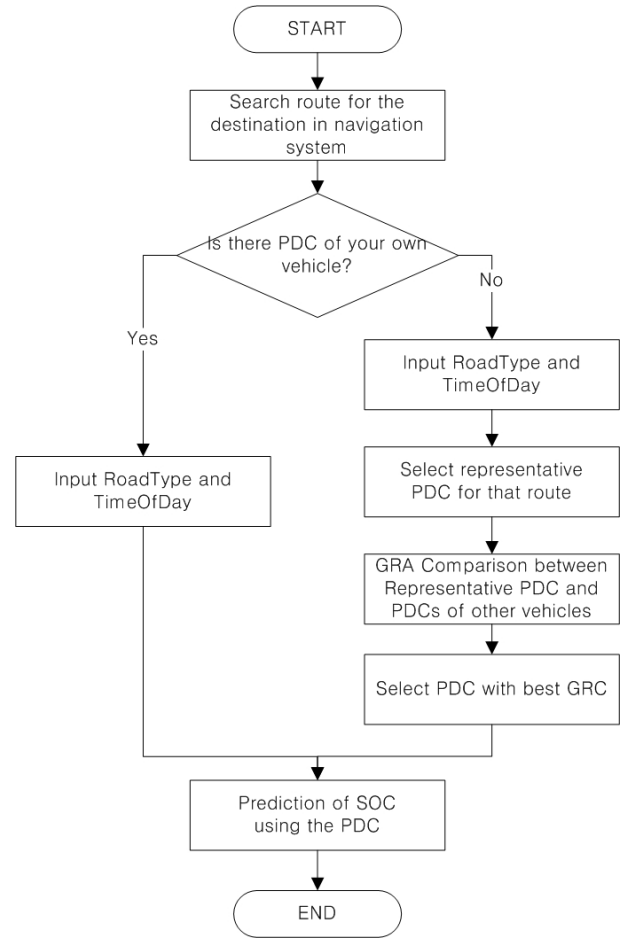
1: data ← load(PDC)
2: time_data ← data[:, 0]
3: velocity_data ← data[:, 1]
4: acc_data ← data[:, 2]
5: vel_acc_phase ← []
6: acc_acc_phase ← []
7: vel_dec_phase ← []
8: dec_dec_phase ← []
9: idx ← 0
10: for row in velocity_data do
11:   if acc_data[idx] > 0 then
12:     vel_acc_phase.append(velocity_data[idx])
13:   else
14:     vel_dec_phase.append(velocity_data[idx])
15:   end if
16:   idx ← idx + 1
17: end for
18: idx ← 0
19: for row in acc_data do
20:   if acc_data[idx] > 0 then
21:     acc_acc_phase.append(acc_data[idx])
22:   else
23:     dec_dec_phase.append(acc_data[idx])
24:   end if
25:   idx ← idx + 1
26: end for
27: avg_vel_acc_phase ← compute_avg(vel_acc_phase)
28: avg_vel_dec_phase ← compute_avg(vel_dec_phase)
29: avg_acc_acc_phase ← compute_avg(acc_acc_phase)
30: avg_dec_dec_phase ← compute_avg(dec_dec_phase)
Output: avg_vel_acc_phase, avg_vel_dec_phase,
          avg_acc_acc_phase, avg_dec_dec_phase

```

For data-driven methods, the performance of a DNN model in the machine learning (ML) module depends on the setting of hyperparameters. Thus, we used the VED dataset and 5-fold cross-validation (CV) to perform hyperparameter tuning.

Our DNN model was implemented through the Python library tensorflow.keras. For the optimizer and the activation function, respectively, 'Adam' and 'relu' were used and the three hyperparameters tuned were number of hidden layers, number of hidden units per layer and number of epochs. Additionally, as can be seen in Algorithm 1, two different DNNs are trained and saved, one for the PC data and another for the PR data. Thus, hyperparameter tuning must be done separately for each model.

As we can see in Algorithm1, PC data and PR data can be used separately to train and save a model as a file. As each DNN model is trained separately for PC data and PR data, the hyperparameter of the model for each data needs to be tested separately.

**FIGURE 6.** SOC prediction through PDC selection.

Tables 3 and 4 are the results of 5-fold cross-validation with PC data and PR data, respectively. From Table 3, it can be seen that the average MAPE decreases as the number of hidden layers increases and the number of neurons increases. However, it has a minimum value of 10.46% when the number of neurons and epochs are, respectively, 256 and 30000, and then increases again. Therefore, the optimal hyperparameter settings of the DNN model for PC data are 2 hidden layers, 256 neurons, and 30000 epochs. Next, as in Table 4, the MAPE value decreased as the number of neurons increased and as there were more hidden layers. The minimum MAPE value of 8.88% was achieved there were 64 neurons and 30000 epochs. Therefore, the optimal parameters of the DNN model for PR data are 2 hidden layers, 64 neurons, and 30000 Epochs.

B. PARAMETERS SETTING FOR SOC PREDICTION IN BSP MODULE OF ODC

Parameters for predicting battery SOC may be obtained through battery data. Using the quadratic equation of formula (9), the initial voltage (E) and internal resistance (R) values can be inferred by using two different SOC values of the battery data. Then, based on these values, the current can be computed. Particularly, the consumption current and

TABLE 3. 5-Fold cross validation result (PC dataset).

Hidden Layer Num	Neuron number	Epoch	Average MAPE
1	32	10000	35.30
		20000	36.42
		30000	29.09
	64	10000	31.80
		20000	30.69
		30000	40.25
	128	10000	33.48
		20000	26.81
		30000	28.99
2	32	10000	30.18
		20000	45.74
		30000	29.13
	64	10000	26.31
		20000	25.25
		30000	20.75
	128	20000	16.89
		30000	15.08
		40000	12.05
	256	20000	13.06
		30000	10.46
		40000	13.64

TABLE 4. 5-Fold cross validation result (PR dataset).

Hidden Layer Num	Neuron number	Epoch	Average MAPE
1	32	10000	21.40
		20000	22.01
		30000	26.46
	64	10000	22.88
		20000	23.99
		30000	21.49
	128	10000	24.61
		20000	15.70
		30000	16.94
2	32	10000	21.12
		20000	17.08
		30000	13.67
	64	10000	21.64
		20000	12.06
		30000	8.88
	128	20000	9.07
		30000	9.63
		40000	12.21

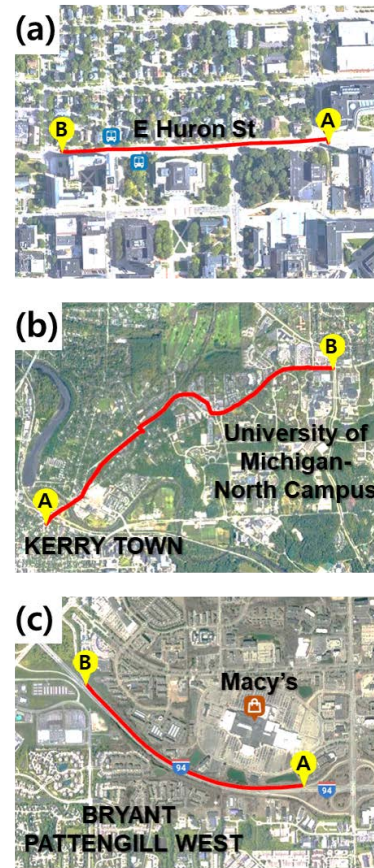
charging current can be obtained through the equations (9) and (10).

C. ENERGY CONSUMPTION PREDICTION RESULTS USING VED

In this paper, to evaluate the performance of the proposed algorithm, tests were conducted based on the public VED of the University of Michigan described in section II.

In addition, since power consumption of EV can vary depending on RoadTypes, the proposed algorithm was evaluated using driving data from each region by dividing it into three RoadTypes as shown in Figure 7 with reference to existing papers and GPS information [28].

The test was conducted by dividing the origin-destination route searched by the navigation system into two scenarios as follows.

**FIGURE 7.** (a) Downtown (b) Urban road (c) Highway region of VED by Googlemap.

- 1) Scenario 1: In case there is driving data of the driver's vehicle for origin-destination route in ODC
- 2) Scenario 2: In case there is no driving data of the driver's vehicle for origin-destination route in ODC

We selected the OBD data set from VED as shown in Table 5 for the test for scenario 1. The reason that only Veh id 455 and Veh id 10 were selected from among the numerous Veh ids of VED is because this vehicle is an EV present in the actual VED.

Furthermore, as mentioned in the text, each CAEV uses its own Veh id to request its own driving and battery information from the VED OBD data. Then, this data is used to train the two ML models related to the PC and PR data.

If the current vehicle has a history of driving the route from the origin to the destination at least once, the battery SOC can be predicted by using the previous personal driving cycle under the assumption that the same driver is driving.

Although the driving cycle of the starting point to the destination is the same, the power used for other things, that is, ambient temperature, air conditioner, heater, etc., auxiliary power consumption, may be different.

Therefore, in this paper, battery SOC prediction test was conducted under the assumption that all other conditions except driving data are the same as the most recent driving information.

TABLE 5. Test environment setting for scenario1.

Road Type	Driver's Veh id VED	Test trip in VED
Downtown	455	1221
Urban road	10	3025
Highway	455	1593

Table 6 and Figure 8 show the accuracy of battery consumption current when using several different ML models and predicting them in the order shown in Figure 5.

As already explained in section III, we first used ML algorithms to predict the power consumed and regenerated, and then obtained the current discharged and charged from the battery. And, finally, the final SOC was predicted based on this battery current.

Also, in this paper, multi-variable linear regression (MLR), deep neural network (DNN) and random forest (RF) models are used for ML learning and prediction.

In order to evaluate the accuracy of power consumption and SOC prediction through ML, we first computed the accuracy of the consumption and the regenerative braking currents by the equations (30) and (31), respectively. Then, we used the accuracy equation (29), which is the average between the two mentioned currents. The equations were designed as such because the consumption and the regenerative braking currents are predicted by two different models, as explained in algorithms (1) and (2), whose outputs are summed through equation 11 to compute the total current.

$$\text{Accuracy}_{\text{total}}(\%) = \frac{\text{Accuracy}_{pc} + \text{Accuracy}_{pr}}{2} \quad (29)$$

$$\text{Accuracy}_{pc}(\%) = 100 - \left| \frac{\text{Pred}_{pc} - \text{True}_{pc}}{\text{True}_{pc}} \right| \quad (30)$$

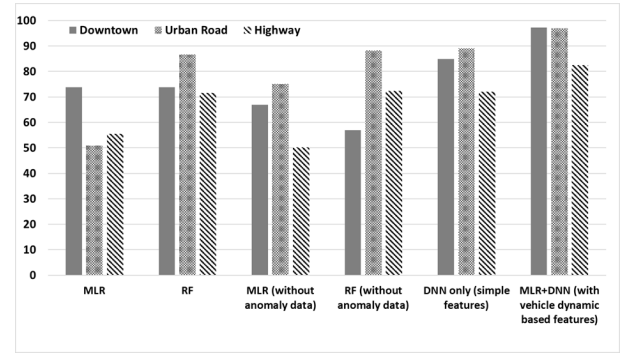
$$\text{Accuracy}_{pr}(\%) = 100 - \left| \frac{\text{Pred}_{pr} - \text{True}_{pr}}{\text{True}_{pr}} \right| \quad (31)$$

The first two ML models MLR and RF from the left in table 6 are the results of prediction with data containing Anomaly Data. After removing the anomalous data, RF shows high accuracy in Urban Road and Highway areas, and almost the same accuracy as MLR in Downtown.

The next two models are the same MLR and RF models, but are the results of learning by removing the anomalous data through Algorithm 1. The predictions made by MLR and RF are only high for Urban Road and, in the case of RF, it can be seen that it has almost the same accuracy as RF with anomalous data. We can conclude that caution is needed because removing anomalous data may not be very helpful if the data does not have a strong linear relationship.

In the case of DNN only, it can be seen that, in most cases, its performance is better than that of MLR and RF, and only in the case of RF without anomalous data on the highway both models have similar performance.

Next, the predicted result with DNN Only is the result of using only DNN without preprocessing using seven basic

**FIGURE 8. Accuracy of energy consumption prediction with different ML models and RoadTypes.**

features: v, a, oat (outside air temperature), heater power, and aircon power in driving data, RoadType, TimeOfDay.

From Figures 8 and 9, it can be seen that the model MLR+DNN had the best performance among all models. This is because this model adds features by considering the vehicle dynamic formula as shown in Table 2. The reason why the results on the highway have poor performance compared to the downtown and urban road areas is presumed to be that the actual vehicle with Veh id 455 is not sufficiently trained because there is almost no data related to driving on the highway.

The accuracy on the highway is relatively low, but it is over 80%, and the average accuracy in all road types shows a high accuracy of over 90%. It can be seen from Figure 9 that this shows about 10% better performance than DNN only, which had the second-best performance. Therefore, it can be seen that the most suitable model to use in the ML module of ODC is MLR+DNN.

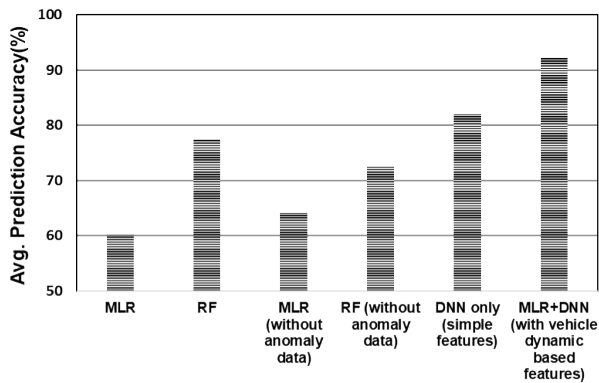
Next, to test the second scenario, the OBD data set was selected from VED as shown in Table 7. Also, in the case of PDC of another vehicle, it was arbitrarily selected and extracted from trip data having GPS information for the routes as shown in Figure 7 among the routes traveled by each Veh id.

In addition, after selecting one PDC by calculating the similarity between PDCs through the DTW and GRA methods, in order to compare the prediction accuracy of actual power consumption, a total of 15 paths (5 for each RoadType) were randomly selected. At this time, the selected PDC was arbitrarily selected from several different vehicles as shown in Table 7 to select that of a vehicle different from the own vehicle, and all PDCs are not the same.

In Table 8, when the power consumption of the corresponding path is predicted for each PDC, the PDCs are numbered so that they are sorted from highest to lowest accuracy. That is, in each RoadType, the PDCs with the highest accuracy are PDC1, PDC6, and PDC11, and the PDCs with the lowest accuracy are PDC5, PDC10, and PDC15. In addition, in order to calculate the DTW cost, we used the PDC value (time versus speed value) stored as a csv file as input to the Matlab's dtw function. Currently, the two PDCs entering the

TABLE 6. Battery SOC prediction result using ML prediction and battery data driven prediction (based self driving cycle).

Road Type	ML algorithm	Battery consumption current		Battery regenerative current		Accuracy	Battery SOC	
		True	Predicted	True	Predicted		True	Predicted
Downtown	MLR	-1114.2	-613.14	317.15	340.5	73.83	53.293	53.613
	RF	-1114.2	-723.03	317.15	372.18	73.77	53.293	53.565
	MLR (w/o anomaly data)	-1114.2	-796.51	317.15	437.01	66.84	53.293	53.560
	RF (w/o anomaly data)	-1114.2	-853.38	317.15	515.7	56.99	53.293	53.573
	DNN only (simple features)	-1114.2	-971.18	317.15	261.57	84.81	53.293	53.346
	MLR + DNN (w/ VD features)	-1114.2	-1098.53	317.15	330.63	97.17	53.293	53.310
Urban Road	MLR	-8535.9	-5064.05	1937.45	822.83	50.90	81.463	83.336
	RF	-8535.9	-7312.21	1937.45	1700.83	86.72	81.463	82.247
	MLR (w/o anomaly data)	-8535.9	-6564.17	1937.45	1416.96	75.01	81.463	82.616
	RF (w/o anomaly data)	-8535.9	-7913.65	1937.45	1624.22	88.27	81.463	81.708
	DNN only (simple features)	-8535.9	-8058.89	1937.45	1621.53	89.05	81.463	81.591
	MLR+DNN (w/ VD features)	-8535.9	-8217.13	1937.45	1890.98	96.93	81.463	81.679
Highway	MLR	-3019.6	-2576.95	158.1	275.49	55.54	75.732	76.185
	RF	-3019.6	-2636.82	158.1	228.14	71.51	75.732	76.098
	MLR (w/o anomaly data)	-3019.6	-2604.01	158.1	293.68	50.24	75.732	76.259
	RF (w/o anomaly data)	-3019.6	-2645.9	158.1	226.1	72.30	75.732	76.170
	DNN only (simple features)	-3019.6	-2570.76	158.1	223.07	72.02	75.732	76.293
	MLR+DNN (w/ VD features)	-3019.6	-2741.71	158.1	199.07	82.44	75.732	76.091

**FIGURE 9.** Average energy consumption prediction accuracy of different ML models.

input are PDCs of other vehicles to be compared with their representative PDCs. To calculate GRC by the GRA method, as described in the section II, the representative PDC and all 5 PDCs were used. In order to implement this, a program that receives a csv file as input and calculates GRC was written and tested as a Python program. In addition, $GW = \{0.5, 0.5\}$ and $LW = \{0.5, 0.5\}$ were used to apply the hierarchy-based weights as shown in Figure 2. Thus, the final weights for the four factors become $W = \{0.25, 0.25, 0.25, 0.25\}$.

In addition, the lower the DTW cost, the higher the similarity, and the larger the GRC, the higher the similarity. If the DTW method is used to select a path with a high similarity, the PDC with the smallest value is selected after calculating the DTW cost. But, in this paper, the power consumption was predicted by selecting the PDC with the largest GRC value.

Figure 10 is a graphical representation of the data in Table 8. First, in the graphs of (a) to (a-2), the DTW cost values for five different PDCs in Downtown, Urbanroad, and

TABLE 7. Test environment setting for scenario2.

Road Type	Driver's Veh id in VED	Others' Veh id in VED
Downtown	455	10, 11, 371
Urban road	10	11, 371, 417
Highway	455	388, 398, 431, 449

Highway are expressed as bar graphs, and in the case of power consumption prediction accuracy, it is expressed as a line graph. In the case of (a), it is good that the one with the smallest DTW cost for PDC1 shows the highest accuracy, but in the case of PDC3, the DTW cost has the relatively largest value but the accuracy has a medium value. In the case of (a-1), it was confirmed that PDC4 had the smallest DTW cost, but the accuracy was the fourth. In the case of (a-2), it was confirmed that PDC13 had the lowest DTW cost, but had a low level of accuracy.

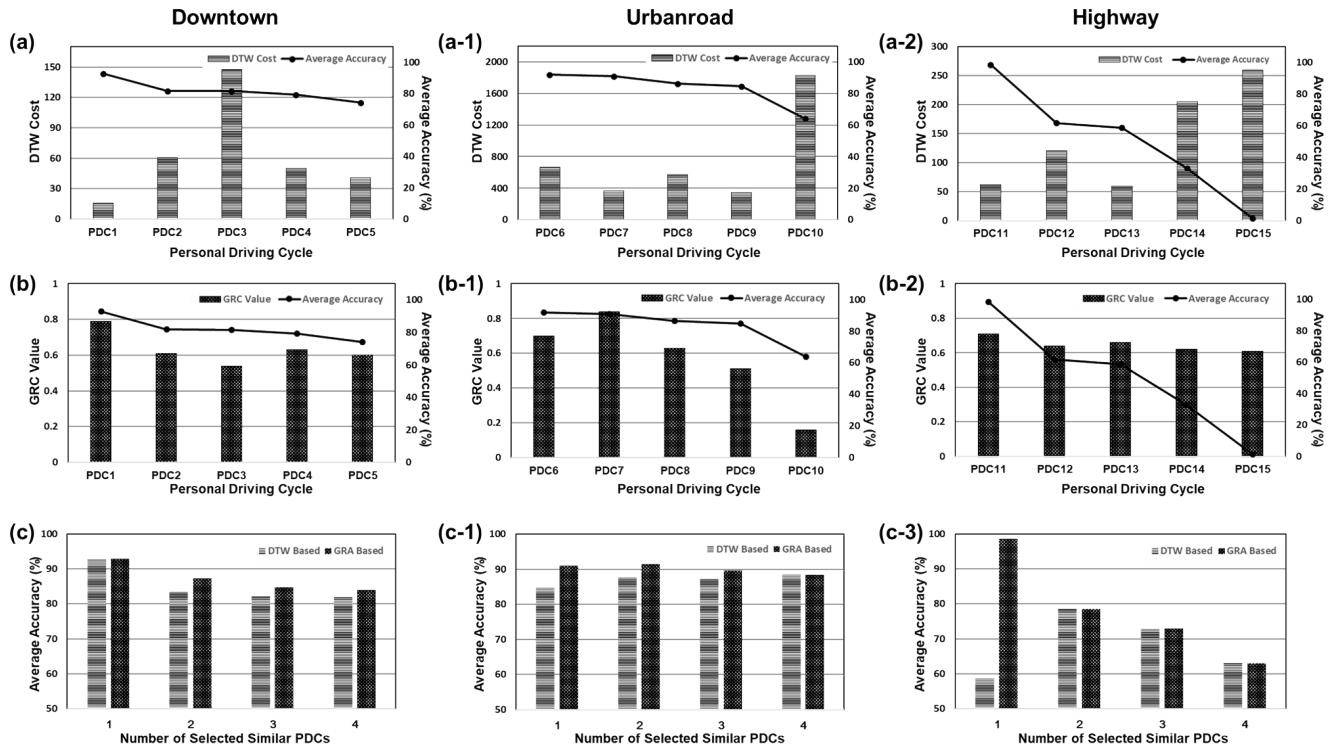
Next, in the graphs of (b) to (b-2), the GRC values for five different PDCs in Downtown, Urbanroad, and Highway are displayed as bar graphs, and in the case of power consumption prediction accuracy, (a) to (a-2) and expressed as a line graph. In the case of (b), it can be confirmed that the PDC1 has the largest GRC and the greatest prediction accuracy. In the case of (b-1), PDC7 has the largest GRC, and the prediction accuracy has the second largest value, and it can be confirmed that this prediction value's accuracy is also more than 90%.

In the case of (b-2), it can be seen that the prediction accuracy of PDC11 is the highest, and the GRC of the actual PDC11 is also the largest, and in the case of PDC15, the prediction accuracy is about 1% and the GRC is the smallest among all paths.

Lastly, the graphs of (c) to (c-2) show 1, 2, and 3 PDCs, respectively, which are the most similar to the representative PDC of the vehicle among the 5 PDCs for each RoadType

TABLE 8. Test results of PDC selection according to the outputs of DTW and GRA.

PDC	Downtown			PDC	Urban road			PDC	Highway		
	Accuracy	DTW cost	GRC		Accuracy	DTW cost	GRC		Accuracy	DTW cost	GRC
PDC1	92.862	15.74	0.79	PDC6	91.8323	663.81	0.7	PDC11	98.56	62.128	0.71
PDC2	81.83	60.49	0.61	PDC7	90.9166	363.68	0.84	PDC12	61.64	120.65	0.64
PDC3	81.71	147.32	0.54	PDC8	86.3459	572.34	0.63	PDC13	58.63	59.64	0.66
PDC4	79.42	50.38	0.63	PDC9	84.7274	347.93	0.51	PDC14	32.85	205.31	0.62
PDC5	74.22	40.75	0.6	PDC10	63.8201	1831.18	0.16	PDC15	1.28	259.63	0.61

**FIGURE 10.** Energy consumption prediction results with PDC selection using DTW and GRA.

(Downtown, Urbanroad, and Highway). It shows the average accuracy for each of them, when up to 4 are selected.

The accuracy (%) calculation when n PDCs are selected is expressed by equation (32).

$$Avg.Acc_{DTW/GRA}^n = \frac{\sum_{i=1}^n P_{DTW/GRA}^{ith}}{n} \quad (32)$$

In equation (32), $P_{DTW/GRA}^{ith}$ is the DTW or GRA algorithm that has the i -th highest prediction accuracy. For example, in Downtown, when two high-accuracy PDCs are selected by DTW, the average accuracy can be expressed as $Avg.Acc_{DTW}^2$. The result is 83.54%, which is the average of 92.86% and 74.22%.

In the case of (c), since both DTW and GRA select the same path when selecting the highest accuracy, it can be seen that the GRA method has a slightly higher average accuracy except for the case where the first one is selected. In the case of Urbanroad, it is better to use GRA in all cases from the first one to the third, and it can be seen that the performance

of DTW and GRA is the same for the fourth case. Also, in the case of Highway, it can be seen that when only one PDC is selected, the GRA is significantly higher than that of the DTW, and the average accuracy is the same as in the DTW method when two or more PDCs are selected.

Next, Figure 11 shows the average accuracy when the most similar route is selected from the three RoadTypes, Downtown/Urbanroad/Highway. While GRA has more than 90% prediction accuracy, DTW method is less than 80% because the path with the highest prediction accuracy may not be selected when using the DTW method.

Finally, Figure 12 shows the absolute value of the average correlation coefficient when the correlation values between the accuracy and each algorithm are evaluated according to each RoadType. The smaller the DTW cost, the higher the prediction accuracy, so it has a negative correlation, and the larger the GRC, the higher the accuracy, so it has a positive correlation, and the absolute value of the correlation coefficient has a value between 0 and 1. It can be seen that GRA

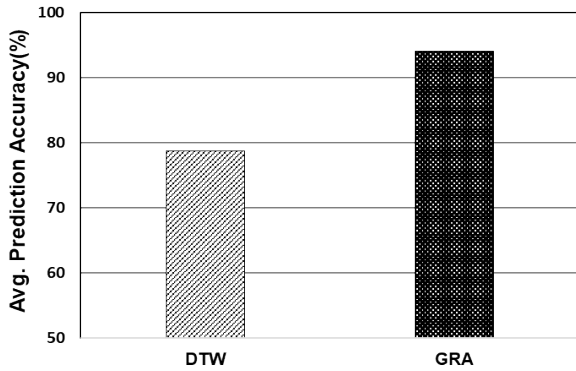


FIGURE 11. Average energy consumption prediction accuracy with DTW and GRA.

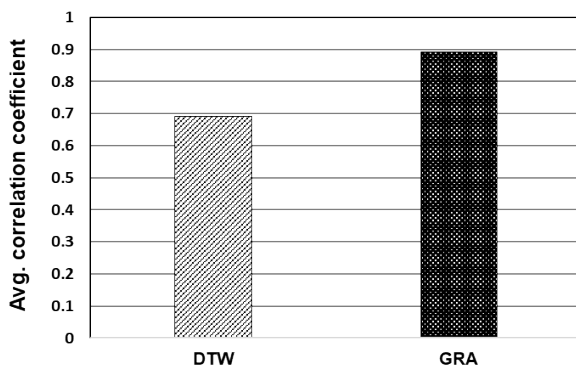


FIGURE 12. Average correlation coefficient between DTW/GRA and energy consumption prediction accuracy.

has a 90% correlation, whereas DTW has a correlation value of about 70%. From this, it can be seen that the power consumption can be predicted more accurately when the power consumption is predicted by selecting a PDC through the GRA method.

V. CONCLUSION

In the case of CAEV, the power consumption of the battery is an important factor in driving. In this paper, a CAEV model to collect OBD data from CAEV, ODC to manage OBD data, and a message exchange sequence is proposed for accurate power consumption prediction based on data. Next, an MLR+DNN model operating in the ML module of ODC is proposed. Additionally, in the case of a new route that has never been driven, a method for selecting PDC and predicting power consumption is presented as a hierarchical GRA method using speed and acceleration information.

The experimental results confirmed that the proposed MLR+DNN model could predict battery current consumption with an average accuracy of 92% for the route that CAEV has traveled before. In the case of a route that has never been traveled before, we tested the prediction of power consumption with 5 PDCs of different vehicles for each road type

(Downtown, Urban, and Highway). Furthermore, an average accuracy of 94% was obtained in selecting the most similar personal driving cycle (PDC) using hierarchical grey relational analysis (GRA), which is about 15% higher than the accuracy obtained by using the dynamic time warping (DTW) method.

In this study, the GRA method, which applies the weight of the hierarchical structure composed of four factors, is used to select the most similar PDC in power consumption. We used the static weights in this study. Therefore, in the future, adding critical factors to power consumption prediction and a study for more accurate power consumption prediction by dynamically allocating weights through ML or reinforcement learning (RL) can be conducted.

REFERENCES

- [1] E. Uhlemann, "Introducing connected vehicles [connected vehicles]," *IEEE Veh. Technol. Mag.*, vol. 10, no. 1, pp. 23–31, Mar. 2015, doi: [10.1109/MVT.2015.2390920](https://doi.org/10.1109/MVT.2015.2390920).
- [2] A. A. Alkheir, M. Aloqaily, and H. T. Mouftah, "Connected and autonomous electric vehicles (CAEVs)," *IT Prof.*, vol. 20, no. 6, pp. 54–61, Nov./Dec. 2018, doi: [10.1109/MITP.2018.2876977](https://doi.org/10.1109/MITP.2018.2876977).
- [3] I. W. Damaj, D. K. Serhal, L. A. Hamandi, R. N. Zantout, and H. T. Mouftah, "Connected and autonomous electric vehicles: Quality of experience survey and taxonomy," *Veh. Commun.*, vol. 28, Apr. 2021, Art. no. 100312, doi: [10.1016/j.vehcom.2020.100312](https://doi.org/10.1016/j.vehcom.2020.100312).
- [4] H. Yu, F. Cheli, and F. Castelli-Dezza, "Optimal design and control of 4-IWD electric vehicles based on a 14-DOF vehicle model," *IEEE Trans. Veh. Technol.*, vol. 67, no. 11, pp. 10457–10469, Nov. 2018.
- [5] Q. Ye and W. Zhuang, "Distributed and adaptive medium access control for Internet-of-Things-enabled mobile networks," *IEEE Internet Things J.*, vol. 4, no. 2, pp. 446–460, Apr. 2017.
- [6] P. W. Shaikh and H. T. Mouftah, "Intelligent charging infrastructure design for connected and autonomous electric vehicles in smart cities," in *Proc. IFIP/IEEE Int. Symp. Integr. Netw. Manage. (IM)*, May 2021, pp. 992–997.
- [7] O. Egbue and S. Long, "Barriers to widespread adoption of electric vehicles: An analysis of consumer attitudes and perceptions," *Energy Policy*, vol. 48, pp. 717–729, Sep. 2012, doi: [10.1016/j.enpol.2012.06.009](https://doi.org/10.1016/j.enpol.2012.06.009).
- [8] J. Deng, C. Bae, A. Denlinger, and T. Miller, "Electric vehicles batteries: Requirements and challenges," *Joule*, vol. 4, no. 3, pp. 511–515, Mar. 2020.
- [9] G. De Nunzio and L. Thibault, "Energy-optimal driving range prediction for electric vehicles," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2017, pp. 1608–1613, doi: [10.1109/IVS.2017.7995939](https://doi.org/10.1109/IVS.2017.7995939).
- [10] Z. Wang, X.-H. Wang, L.-Z. Wang, X.-F. Hu, and W.-H. Fan, "Research on electric vehicle (EV) driving range prediction method based on PSO-LSSVM," in *Proc. IEEE Int. Conf. Prognostics Health Manage. (ICPHM)*, Jun. 2017, pp. 260–265, doi: [10.1109/ICPHM.2017.7998338](https://doi.org/10.1109/ICPHM.2017.7998338).
- [11] H. A. Yavasoglu, Y. E. Tetik, and K. Gokce, "Implementation of machine learning based real time range estimation method without destination knowledge for BEVs," *Energy*, vol. 172, pp. 1179–1186, Apr. 2019, doi: [10.1016/j.energy.2019.02.032](https://doi.org/10.1016/j.energy.2019.02.032).
- [12] B. Varga, A. Sagoian, and F. Mariasiu, "Prediction of electric vehicle range: A comprehensive review of current issues and challenges," *Energies*, vol. 12, no. 5, p. 946, Mar. 2019, doi: [10.3390/en12050946](https://doi.org/10.3390/en12050946).
- [13] L. Zhao, W. Yao, Y. Wang, and J. Hu, "Machine learning-based method for remaining range prediction of electric vehicles," *IEEE Access*, vol. 8, pp. 212423–212441, 2020, doi: [10.1109/ACCESS.2020.3039815](https://doi.org/10.1109/ACCESS.2020.3039815).
- [14] S. Sun, J. Zhang, J. Bi, and Y. Wang, "A machine learning method for predicting driving range of battery electric vehicles," *J. Adv. Transp.*, vol. 2019, pp. 1–14, Jan. 2019, doi: [10.1155/2019/4109148](https://doi.org/10.1155/2019/4109148).
- [15] J.-Y. Yang, L.-D. Chou, and Y.-J. Chang, "Electric-vehicle navigation system based on power consumption," *IEEE Trans. Veh. Technol.*, vol. 65, no. 8, pp. 5930–5943, Aug. 2016, doi: [10.1109/TVT.2015.2477369](https://doi.org/10.1109/TVT.2015.2477369).

- [16] D. Tanaka, T. Ashida, and S. Minami, "An analytical method of EV velocity profile determination from the power consumption of electric vehicles," in *Proc. IEEE Vehicle Power Propuls. Conf.*, Sep. 2008, pp. 1–3, doi: [10.1109/VPPC.2008.4677742](https://doi.org/10.1109/VPPC.2008.4677742).
- [17] E. Yao, Z. Yang, Y. Song, and T. Zuo, "Comparison of electric vehicle's energy consumption factors for different road types," *Discrete Dyn. Nature Soc.*, vol. 2013, pp. 1–7, Dec. 2013, doi: [10.1155/2013/328757](https://doi.org/10.1155/2013/328757).
- [18] S. Grubwinkler and M. Lienkamp, "A modular and dynamic approach to predict the energy consumption of electric vehicles," in *Proc. CoFAT*, 2013, pp. 1–10.
- [19] J. Larminie and J. Lowry, *Electric Vehicle Technology Explained*. Hoboken, NJ, USA: Wiley, 2012.
- [20] K. L. Butler, M. Ehsani, and P. Kamath, "A MATLAB-based modeling and simulation package for electric and hybrid electric vehicle design," *IEEE Trans. Veh. Technol.*, vol. 48, no. 6, pp. 1770–1778, Nov. 1999.
- [21] A. Bhatt, "Planning and application of electric vehicle with MATLAB/simulink," in *Proc. IEEE Int. Conf. Power Electron., Drives Energy Syst. (PEDES)*, Dec. 2016, pp. 1–6, doi: [10.1109/PEDES.2016.7914220](https://doi.org/10.1109/PEDES.2016.7914220).
- [22] S. Soyulu, *Electric Vehicles: Modelling and Simulations*. Norderstedt, Germany: BoD-Books on Demand, 2011.
- [23] C.-M. Tseng and C.-K. Chau, "Personalized prediction of vehicle energy consumption based on participatory sensing," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 11, pp. 3103–3113, Nov. 2017.
- [24] K. Vatanparvar, S. Faezi, I. Burago, M. Levorato, and M. A. Al Faruque, "Extended range electric vehicle with driving behavior estimation in energy management," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 2959–2968, May 2019.
- [25] E. Erricson, "Independent driving pattern factors and their influence on fuel-use and exhaust emission factors," *Transp. Res. D, Transp. Environ.*, vol. 6, no. 5, pp. 325–341, Sep. 2001.
- [26] M. Ehsani, Y. Gao, S. E. Gay, and A. Emadi, *Modern Electric Hybrid Electric and Fuel Cell Vehicles*. Boca Raton, FL, USA: CRC Press, 2005.
- [27] C. De Cauwer, W. Verbeke, T. Coosemans, S. Faid, and J. Van Mierlo, "A data-driven method for energy consumption prediction and energy-efficient routing of electric vehicles in real-world conditions," *Energies*, vol. 10, no. 5, p. 608, May 2017. [Online]. Available: <http://www.mdpi.com/1996-1073/10/5/608>
- [28] G. Castignani, R. Frank, and T. Engel, "Driver behavior profiling using smartphones," in *Proc. 16th Int. IEEE Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2013, pp. 552–557, doi: [10.1109/ITSC.2013.6728289](https://doi.org/10.1109/ITSC.2013.6728289).
- [29] G. Castignani, T. Derrmann, R. Frank, and T. Engel, "Driver behavior profiling using smartphones: A low-cost platform for driver monitoring," *IEEE Intell. Transp. Syst. Mag.*, vol. 7, no. 1, pp. 91–102, Spring 2015, doi: [10.1109/ITS.2014.2328673](https://doi.org/10.1109/ITS.2014.2328673).
- [30] R. Araujo, A. Igreja, R. de Castro, and R. E. Araujo, "Driving coach: A smartphone application to evaluate driving efficient patterns," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2012, pp. 1005–1010, doi: [10.1109/IVS.2012.6232304](https://doi.org/10.1109/IVS.2012.6232304).
- [31] H. Eren, S. Makinist, E. Akin, and A. Yilmaz, "Estimating driving behavior by a smartphone," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2012, pp. 234–239, doi: [10.1109/IVS.2012.6232298](https://doi.org/10.1109/IVS.2012.6232298).
- [32] C. Saiprasert, S. Thajchayapong, T. Pholprasit, and C. Tanprasert, "Driver behaviour profiling using smartphone sensory data in a V2I environment," in *Proc. Int. Conf. Connected Vehicles Expo (ICCVE)*, Nov. 2014, pp. 552–557, doi: [10.1109/ICCVE.2014.7297609](https://doi.org/10.1109/ICCVE.2014.7297609).
- [33] H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-26, no. 1, pp. 43–49, Feb. 1978, doi: [10.1109/TASSP.1978.1163055](https://doi.org/10.1109/TASSP.1978.1163055).
- [34] Q. Song and A. Jamalipour, "Network selection in an integrated wireless LAN and UMTS environment using mathematical modeling and computing techniques," *IEEE Wireless Commun.*, vol. 12, no. 3, pp. 42–48, Jun. 2005, doi: [10.1109/MWC.2005.1452853](https://doi.org/10.1109/MWC.2005.1452853).
- [35] J. L. Deng, "Introduction to grey system theory," *J. Grey Syst.*, vol. 1, no. 1, pp. 1–24, 1989.
- [36] G. Oh, D. J. Leblanc, and H. Peng, "Vehicle energy dataset (VED), a large-scale dataset for vehicle energy consumption research," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3302–3312, Apr. 2022, doi: [10.1109/TITS.2020.3035596](https://doi.org/10.1109/TITS.2020.3035596).
- [37] G. M. Platform. Elevation API: Get started. Google. Accessed: Mar. 29, 2022. [Online]. Available: <https://developers.google.com/maps/documentation/elevation/start>



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